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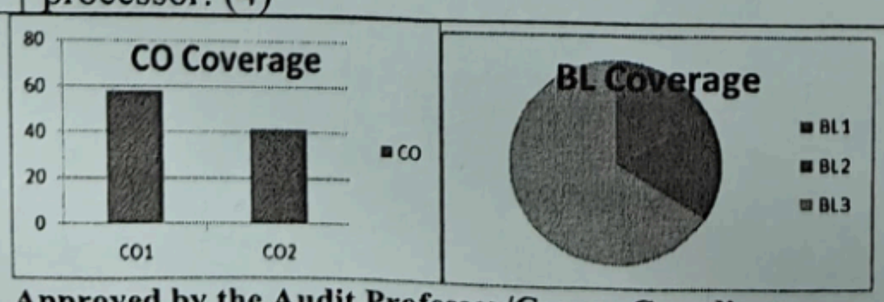
SET D

SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

Academic Year: 2025-26 (ODD)

Test: FT II **Date: 15.09.25**
Course Code & Title: 21CSS201T/Computer Organization and Architecture **Duration:** 2 Hours
Year & Sem: II Year / III Sem **Max. Marks:** 50

	Part – B **(4 x 10 =40 Marks)** **Answer any FOUR Questions**					
11	a) A signed decimal value of -99 needs to be encoded for storage in a digital system that uses two's complement representation. Show how this number would appear in binary, octal, and hexadecimal formats. (4Marks) b) A fractional binary quantity is given as $(1101101.101)_2$. If this value is to be interpreted by different numbering systems, what would be its equivalent in decimal, octal, and hexadecimal? (3 Marks). c) A hexadecimal code $(4F.A)_{16}$ is recorded in a memory register. Determine its corresponding representation in binary and its equivalent decimal value. (3 Marks)	10	3	1	PO1	1.1.2
12	Perform the following operations in binary: i) Find the sum for the given numbers: A = 10110110 ; B = 01101011 ; C = 11001100 ; D = 00110101 ii) Calculate the difference of $(100101)_2 - (10111)_2$ using 2's complement. iii) Find the product of $(1101)_2 \times (111)_2$ iv) Perform division for given binary values: Dividend= $(101100)_2$; Divisor=$(100)_2$. v) Perform BCD addition for the given decimal: 576 + 389. (Each carries 2 marks)	10	3	1	PO1	1.1.2
13	Solve the following Boolean expression using all NAND gates: a) F = A'B + AB' + ABC	10	3	1	PO2	2.2.1
14	Write an Assembly Language Program for the equation: Z=(X+Y)×(W−V)+U; using Zero-Address, One-Address, and Two-Address instruction styles. (3+4+4 marks)	10	2	2	PO2	2.2.2
15	a) Describe the Straight Line Sequencing (SLS) in microprocessor operations. Illustrate its instruction execution cycle and discuss its advantages and limitations. (6) b) Write about the general-purpose registers of 8086 processor. (4)	10	2	2	PO2	2.2.3



Approved by the Audit Professor/Course Coordinator